The future of Commercial Robots

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Course: Intelligent and Autonomous Robots

Master in Robotics and Automation, Carlos III de Madrid University

State-of-the-art of Commercial Robots

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Current state of Commercial Robots

Disruptive techonologies

> *What do we have today?* 🌍

- Artificial Intelligence and Machine Learning
- Computer Vision
- Micro, nano and soft robotics.
- Edge, Cloud and Quantum Computing
- Computer processing
- Advance Material Science

Current state of Commercial Robots

Current potential Challenges

> *Are there political and societal limitations?* 🤔

- Ethical considerations
- Privacy and security
- Job market transformation
- Regulatory frameworks
- Societal acceptance

5-year Horizon (2029)

- Increasing AI-driven personalization
- Enhanced natural language processing

5-year Horizon (2029)

1. Healthcare and Assistive Robots

Application	Technology	Capabilities	Market_Potential
Surgical Assistants	Micro-robotic Instruments	Precision surgeries, minimal invasiveness	High
Elderly Care Robots	Soft Exoskeletons	Personal assistance, mobility support	Very High
Rehabilitation Robots	Adaptive Neural Interfaces	Personalized therapy, movement retraining	High

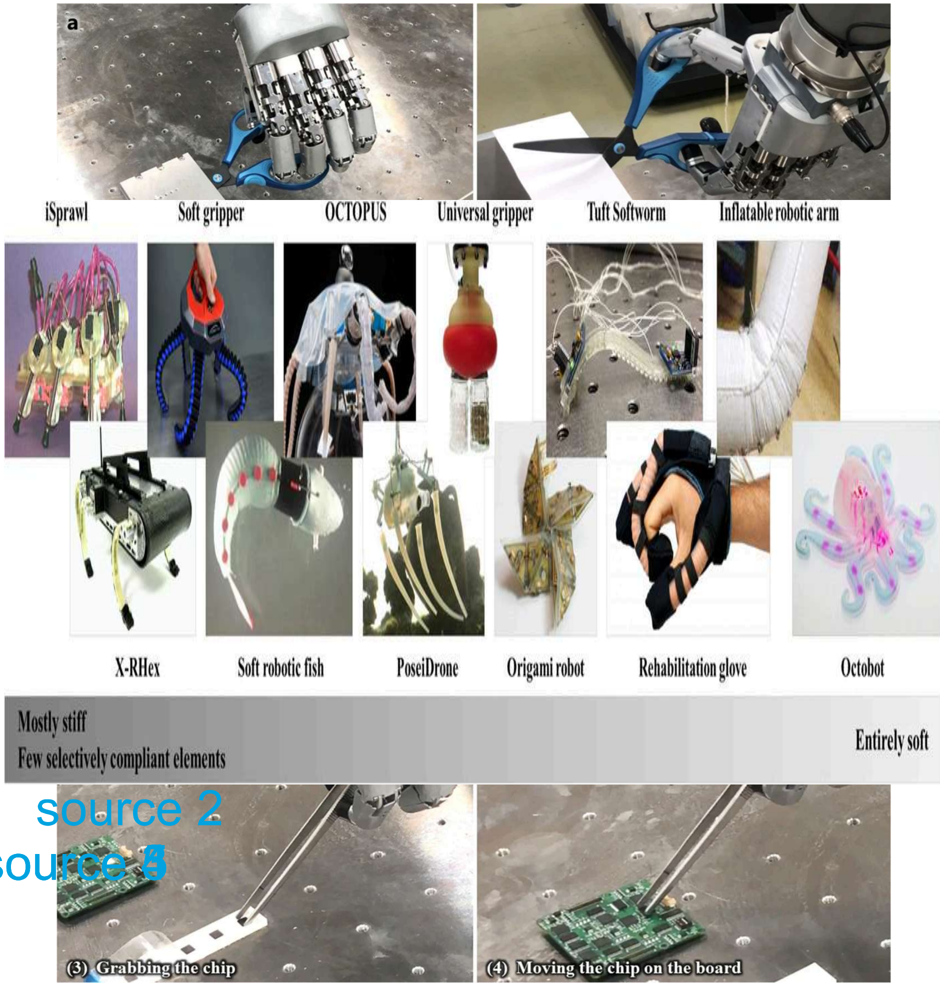
5-year Horizon (2029):

2. Manufacturing and Industrial Robots

Application	Capabilities	Benefits	Market_Potential
Better collaborative robots	Enhanced safety features and control interfaces	Improved worker safety	Very High
Autonomous bio-inspired robots	Reinforcement learning based on human feedback	Increased efficiency and reduce training costs	Very High
Computer-vision aided robots	Rapid reconfiguration and feature recognition	Autonomous control systems	Very High
AI powered drones and quadcopters systems	Fully autonomous delivery robots and drones	Reduced downtime and optimal delivery performance	High
Soft Robotics in Assembly	Delicate process handling	Precision in delicate tasks	High

5-year Horizon (2029): Examples

Current robots



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Future robots

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10-year Horizon (2034)

- Significant human-robot collaboration
- Focus on augmentation rather than full automation

10-year Horizon (2034)

Consumer and Home Robotics

Robot_Type	Functionality	Unique_Features	Expected_Impact
Multi-purpose Home Assistant	Domestic tasks, emotional support	Advanced contextual understanding	Very High
Personal Healthcare Companion	Health monitoring, early diagnosis	Continuous biometric tracking	High
Educational Robotic Tutors	Personalized learning	Adaptive curriculum, emotional intelligence	High

10-year Horizon (2034): Examples

Current robots

Future robots

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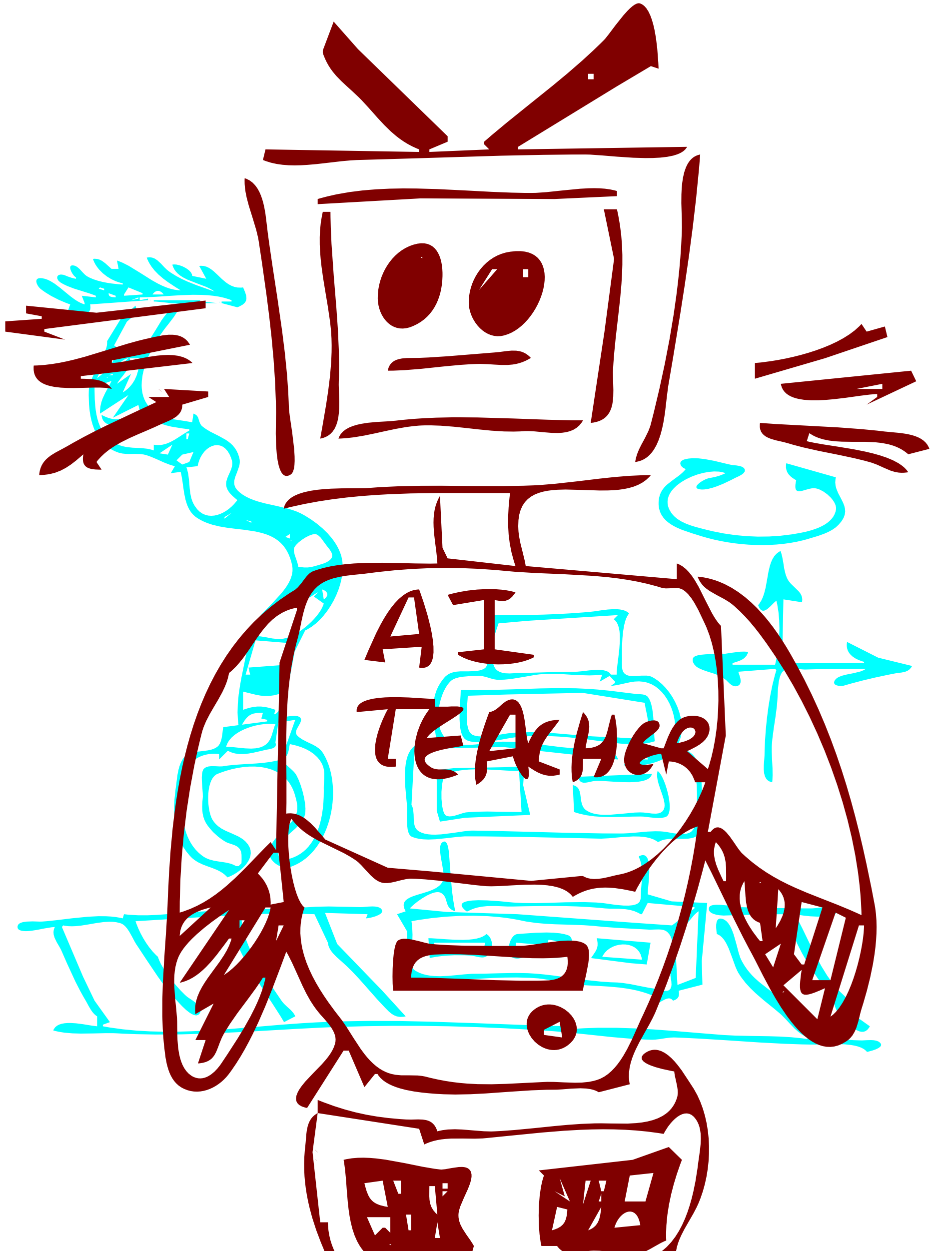


20-year Horizon (2034)

- Highly specialized, purpose-built robotic systems

- Soft Robotics Revolution
- Biomimetic design principles

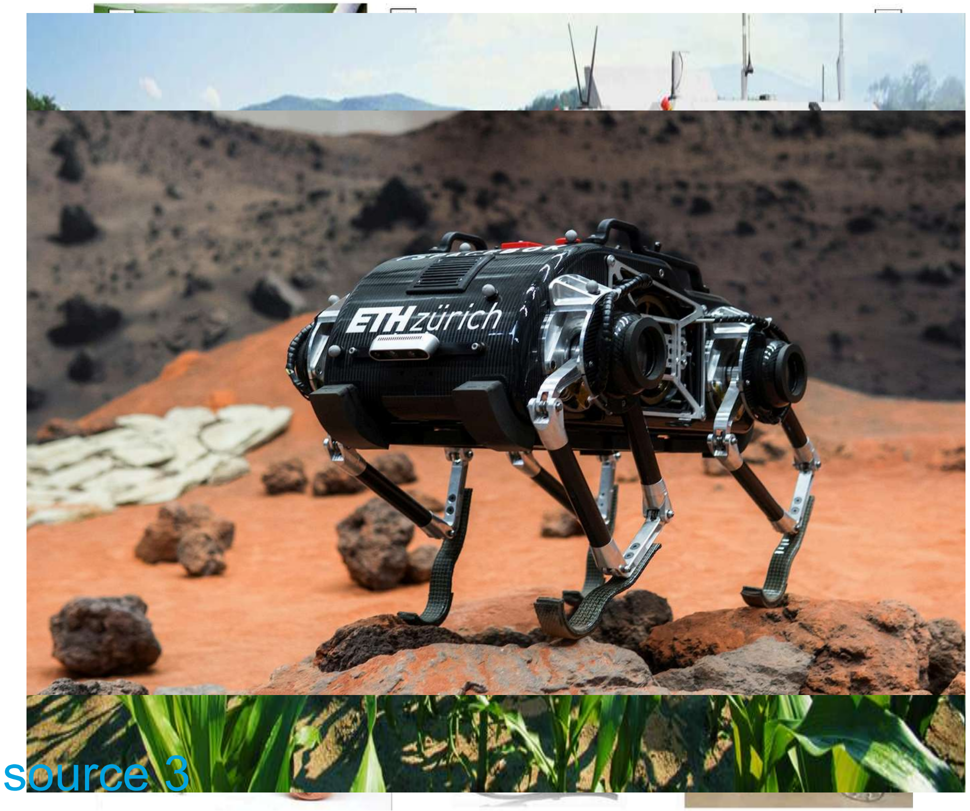
20-year Horizon (2044)

Revolutionary Robots

Domain	Predicted_Capabilities	Enabling_Technologies
Biorobotics	Living tissue integration, artificial tendons manipulation	Synthetic biology, advanced materials
Environmental manipulation	Large-scale ecosystem management: sowing, harvesting, fertilizing	Swarm robotics, nano-scale and precise wide-range sensors
Space Exploration	Adaptive, self-replicating exploration units	Advanced AI, quantum navigation systems and eco-based batteries

20-year Horizon (2044): Examples

Current robots



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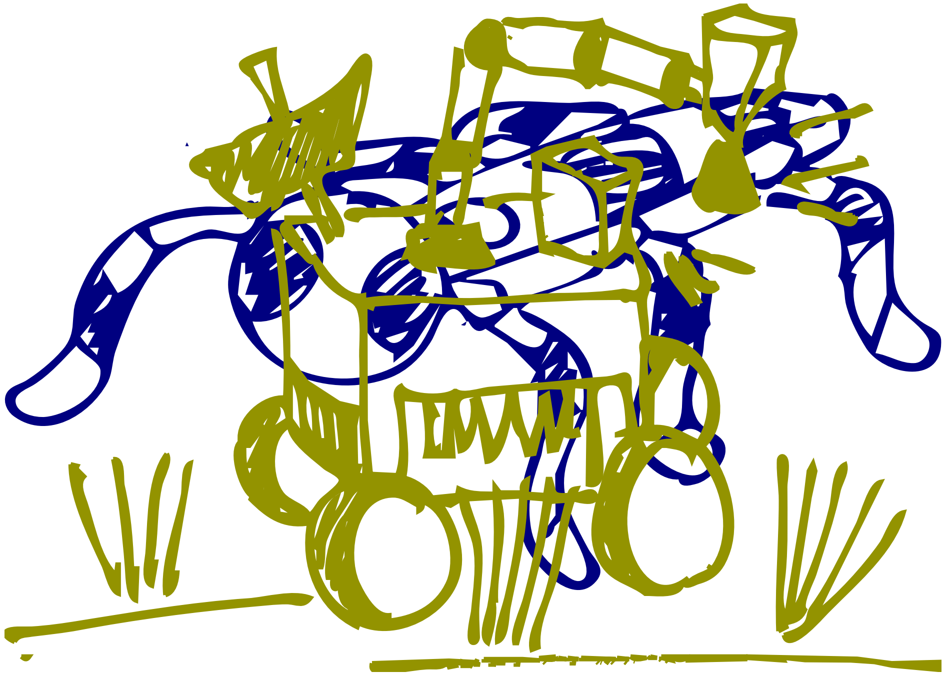
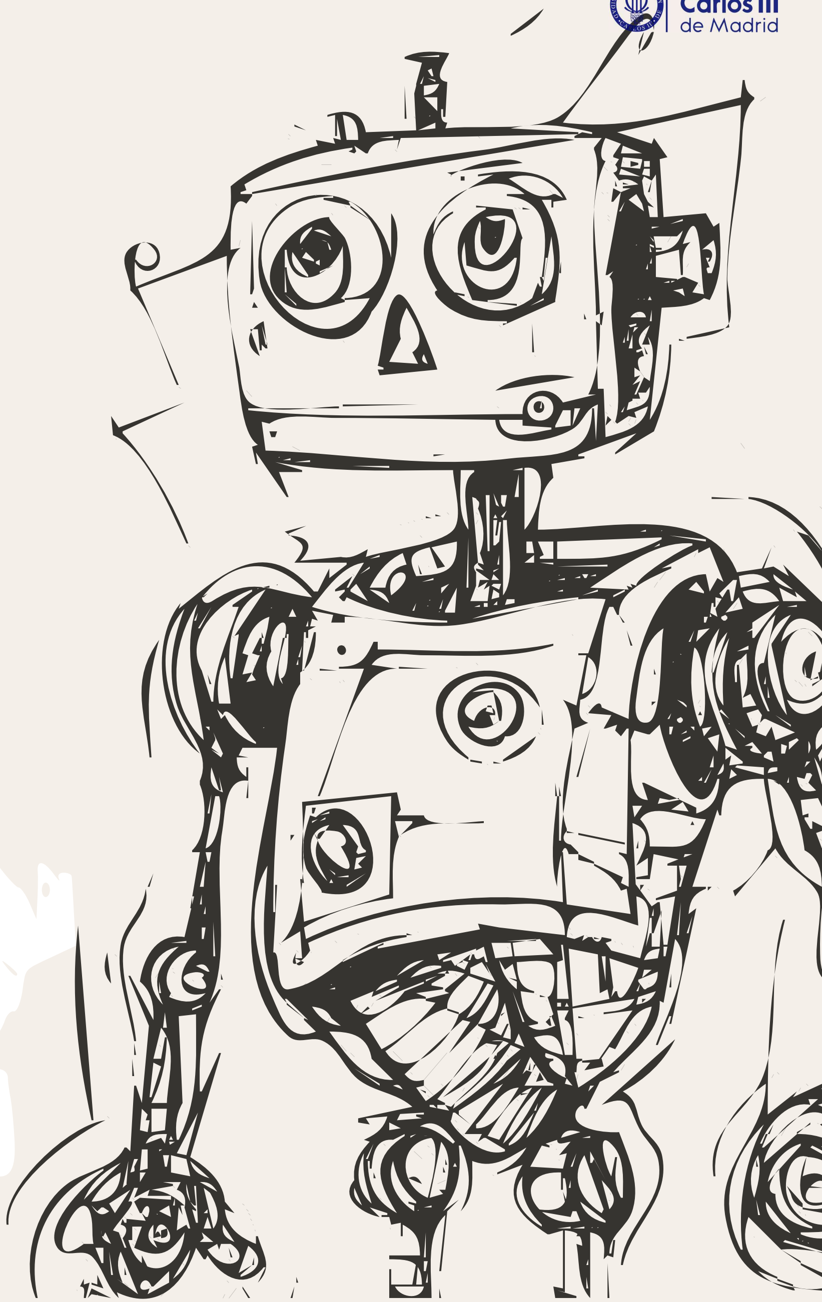
Future robots



Conclusions

- Robots will transform industries and daily life.
- In healthcare, precision surgeries will be enhanced by micro-robotic instruments.
- In the consumer sector, home assistants will become more advanced, offering personalized healthcare monitoring.
- Human-robot collaboration will focus on augmenting human capabilities.
- Revolutionary advancements will include biorobotics with living tissue integration and adaptive robots for space exploration.

Thanks!



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